

This is the most important question you've asked so far because **this is how Company Brain becomes a "Company Brain" instead of just another RAG system.**

The key thing to understand is:

Module 3 does NOT start from scratch. It depends completely on Modules 1 and 2.

Let's follow one document through the entire system.

Company uploads a document

Suppose a company uploads this meeting document.

Meeting Notes

Date: July 25

Alice approved Project Phoenix.

Project Phoenix depends on Legal Approval.

Budget increased to \$2.8M.

Deployment planned for August.

Now let's see what every module does.

Module 1 - Multi-Modal Extraction

Input

Meeting.pdf

Module 1 extracts text.

Meeting Notes

Alice approved Project Phoenix.

Project Phoenix depends on Legal Approval.

Budget increased to \$2.8M.

Nothing intelligent happens here.

It only converts

PDF

↓

Text

Module 2 - Chunking & Embeddings

Module 2 breaks the document into chunks.

Example

Chunk 1

Alice approved Project Phoenix.

Chunk 2

Project Phoenix depends on Legal Approval.

Chunk 3

Budget increased to \$2.8M.

Each chunk gets

- embedding
- metadata
- document id
- page number
- source

Example

Chunk ID: 4812

Document:
Meeting.pdf

Page:

3

Embedding:
[0.27, 0.88, ...]

These are stored in PostgreSQL + pgvector.

Module 2 is finished.

Module 3 Starts Here

Now Module 3 asks

"Can I understand what these chunks mean?"

It starts with Module 2's output.

Chunk

↓

Hybrid Extractor

Why do we need Extractors?

Without an extractor, the AI only sees text.

Alice approved Project Phoenix.

The graph cannot be built directly from this sentence.

First, we must identify:

- People
- Projects
- Companies
- Money
- Dates
- Tasks
- Decisions

That is the job of the extractor.

Hybrid Extractor

Input

Alice approved Project Phoenix.

Budget increased to \$2.8M.

Output

Entities

Alice

Project Phoenix

Facts

Budget = \$2.8M

Relationships

Alice

approved

Project Phoenix

Notice something important.

The graph **does not exist yet**.

We are only extracting structured information.

Why Hybrid?

Imagine these sentences.

Sentence 1

Revenue = \$2.8M

Regex can find

\$

↓

Number

↓

Revenue

No AI required.

Very accurate.

Sentence 2

Alice strongly recommended delaying deployment until legal approval completed.

Regex cannot understand this.

Now the LLM or NLP plugin understands

Decision

↓

Delay Deployment

Therefore

Simple

↓

Regex

Complex

↓

Plugin / LLM

This is why it is called **Hybrid Extraction**.

After Extraction

Now Company Brain has

Entities

Alice

Project Phoenix

Legal Approval

Facts

Budget = \$2.8M

Relationships

Alice approved Project Phoenix

Project Phoenix depends_on Legal Approval

Now we can build the graph.

Entity Resolution

Suppose another document says

MSFT

Another says

Microsoft Corporation

Extractor extracts both.

Now Entity Resolver checks

MSFT

↓

Microsoft Corporation

Same company.

Only one entity is kept.

Now Graph Creation Starts

This is where many people get confused.

A graph has only two things.

Nodes

Edges

Think of Google Maps.

Cities



Nodes

Roads



Edges

Company Brain works exactly the same way.

Step 1 — Create Nodes

Entities become nodes.

Extractor found

Alice

Project Phoenix

Legal Approval

Database

```
graph_nodes
```

Node ID	Name
1	Alice
2	Project Phoenix
3	Legal Approval

These are circles in the graph.

```
(Alice)
```

```
(Project Phoenix)
```

```
(Legal Approval)
```

Still disconnected.

Step 2 — Create Edges

Extractor also found

```
Alice approved Project Phoenix
```

Now create edge

```
Alice
```

```
————approved————>
```

```
Project Phoenix
```

Database

```
graph_edges
```

From	Relation	To
Alice	approved	Project Phoenix

Extractor also found

```
Project Phoenix depends_on Legal Approval
```

Create another edge.

```
Project Phoenix
```

```
——depends_on——>
```

```
Legal Approval
```

Now graph becomes

```
Alice
```

```
|
```

```
approved
```

```
▼
```

```
Project Phoenix
```

```
|
```

```
depends_on
```

```
▼
```

```
Legal Approval
```

Now Company Brain understands relationships.

Where did the edge come from?

Very important.

Every edge has proof.

Example

```
Alice approved Project Phoenix
```

came from

Meeting.pdf

Page 3

Paragraph 2

Store in

graph_relationship_sources

Later CEO asks

"Who said Alice approved it?"

Company Brain answers

Meeting.pdf

Page 3

Everything is explainable.

Knowledge Reasoning

Now graph exists.

Project Phoenix

↓

depends_on

↓

Legal Approval

Later

Legal Approval changes

Status

Overdue

Reasoner traverses graph.

Legal Approval

↓

Overdue

↓

Project Phoenix blocked

↓

Risk

Nobody wrote

Project Phoenix is at Risk.

Company Brain inferred it.

New fact

Project Phoenix

↓

At Risk

gets stored.

Impact Analysis

Now CEO changes policy.

Remote Work Policy

Graph already exists.

Impact Analyzer walks the graph.

Remote Work Policy

↓

Engineering

↓

VPN

↓

IT

↓

Employees

Now it knows

Changing one policy affects four departments.

Organizational Health

Now Company Brain looks at the whole graph.

Questions

How many projects?

How many owners?

Missing workflows?

Conflicts?

Old documents?

Missing entities?

Suppose

18 projects

↓

5 without owners

Health score decreases.

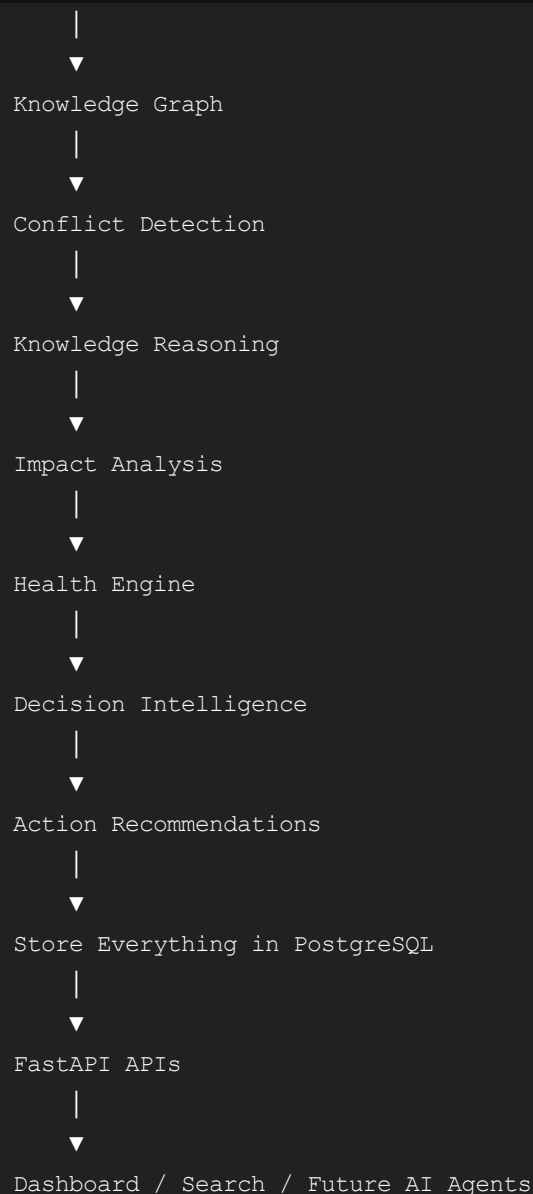
Recommendation

Assign owners

Complete Flow

This is the entire pipeline.





The most important concept

The **extractor** is the bridge between unstructured text and the knowledge graph.

Without extractors, Module 3 only has text and embeddings—it cannot know what should become a node or how nodes should be connected.

The flow is:

1. **Module 1** converts files into text.
2. **Module 2** makes that text searchable using chunks and embeddings.
3. **Module 3 Hybrid Extractor** reads the retrieved text and identifies:
 - **Entities** → become **graph nodes**
 - **Relationships** → become **graph edges**
 - **Facts** → become properties attached to entities or relationships
4. The remaining Module 3 components (entity resolution, reasoning, conflict detection, impact analysis, etc.) enrich and maintain that graph.

That is how Company Brain gradually transforms a collection of documents into a structured representation of how a company actually works.